Multilevel Multimodal CNN–Transformer Fusion Network for Land Use and Land Cover Classification

Guangkuo Xue, Junxiao Zhang, Kexian Sun, Qinglong Shao, Ziqi Zhou, Xiao Zhou, Yuhan Zhang, and Hui Gao

***Abstract*—In remote sensing (RS), convolutional neural networks (CNNs) are well-recognized for their spatial–spectral feature extraction capabilities, whereas vision transformers (ViTs), which leverage attention mechanisms in mining long-range dependencies, have become prominent in land use and land cover (LULC) classification (CLS). As single-modal LULC approaches easily encounter performance bottlenecks, multimodal information fusion methods have gained increasing attention because of their potential to overcome these limitations. However, existing multimodal fusion methods are typically restricted to a single level, that is, they either extract multimodal features from single images or derive single features from multimodal images without fully leveraging the multimodal characteristics at both the image and feature levels. To this end, we propose a novel multilevel multimodal CNN–transformer (MCT) fusion network that simultaneously integrates multimodal information at both the image and feature levels. The MCT network incorporates a multifeature extractor (MFE) to utilize the powerful spatial–spectral information extraction of the CNNs alongside the global representation strengths of transformers. Additionally, a cross-architecture fusion (CAF) module is designed to facilitate a comprehensive fusion of multimodal information across architectures and levels. We conduct extensive experiments on three multimodal benchmark datasets, i.e., the Houston2013 dataset containing hyperspectral (HS) and light detection and ranging (LiDAR) data, the Berlin dataset comprising HS and synthetic aperture radar (SAR) data, and the Augsburg dataset consisting of HS, SAR, and digital surface model (DSM) data, to demonstrate that our proposed method consistently outperforms state-of-the-art (SOTA) methods and to underscore the importance of multimodal information fusion for the advancement of LULC CLS.**

***Index Terms*—Classification (CLS), convolutional neural network (CNN), multilevel multimodal fusion, remote sensing (RS), vision transformer (ViT).**

I. Introduction

As EARTH observation (EO) technologies have advanced rapidly, an abundance of multisource and multimodal remote sensing (RS) data have become readily accessible and have found extensive applications in various fields [1], including urban planning [2], environmental monitoring [3], mineral exploration [4], and object/target detection [5]. In this article, we concentrate specifically on addressing one of the most formidable tasks in land use and land cover (LULC) classification (CLS) using multimodal RS data, which provides valuable insights into changes in land patterns over time, aiding decision-makers in efforts to develop in a sustainable way and effective plan land use.

Given one remotely observed image, the objective of LULC classification is to assign a unique label to each pixel [6], which is similar to the goal of image segmentation tasks in the field of image processing. However, what distinguishes them is the different data sources used. Recent progress in RS technology has significantly increased the accessibility of multisensor data, facilitating various representations of the same geographical region. Depending on the unique characteristics of the sensors involved, the captured data can provide information with different characteristics for the same observed land cover region [7]. Hyperspectral images (HSIs) are highly advantageous in LULC classification because of their ability to capture rich spectral information, allowing for detailed and accurate characterization of land cover types and environmental features. In addition, light detection and ranging, known as LiDAR, employs laser light to measure distances precisely, enabling the creation of detailed 3-D models of the Earth's surface. In contrast, synthetic aperture radar (SAR) uses radar technology to produce high-resolution images, overcoming the limitations imposed by weather conditions and daylight. The digital surface model (DSM) incorporates both topographical features and above-ground structures to represent the Earth's surface, offering a comprehensive depiction of the terrain. Integrating these data with different modalities offering distinctive and complementary information enables the acquisition of comprehensive feature representations. A variety of excellent handcrafted and machine learning-based methods have been proposed for use with the aforementioned multimodal RS data for LULC classification. Typically, these methods comprise two stages. Initially, they reduce the dimensions of the HSIs and extract informative features. Subsequently, spectral classifiers use these features for classification purposes [8], [9]. Unlike HSIs, LiDAR, SAR, and DSM data always serve as complementary information in traditional LULC classification methods [10], [11], [12]. Despite the diversity

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offered by multimodal RS data, these traditional methods still encounter performance bottlenecks as the training size gradually increases and the training set becomes more complex. This limitation may arise from the constrained data fitting and representation capabilities of these traditional methods.

Deep learning (DL) techniques have emerged to meet current needs and have the ability to automatically learn intricate patterns and representations from complex and unstructured data; two DL approaches, convolutional neural networks (CNNs) and transformer-based methods, have gained widespread attention [13], [14]. CNN excels in feature learning by employing convolutional layers that capture hierarchical patterns in spatial data. These layers use filters to detect low-level features, gradually combining them to form higher level representations. For example, Paoletti et al. [15] attempted to construct an efficient spatial–spectral CNN for spatial–spectral classification of hyperspectral (HS) data. Wu et al. [16] developed a CNN-based backbone architecture incorporating a cross-channel reconstruction module to extract CNN-based features from HS and multimodal imagery. Although CNNs demonstrate outstanding performance, their inherent network architecture limits their ability to handle long sequences of spectral signatures effectively. As a consequence, the transformer, which was initially designed for sequence-to-sequence tasks, has shown remarkable adaptability to various domains [14]. It uses self-attention (SA) mechanisms to weigh different parts of the input sequence, allowing it to capture long-range dependencies and relationships within the data. SpectralFormer [17] proposed a cross-layer transformer encoder module that learns spectral representations via adjacent bands, and it is the first transformer explicitly designed for HSI classification. In [18], a deep hierarchical vision transformer (ViT) architecture was proposed to utilize the modeling capability of long-range dependencies and generalizability across different domains of the transformer network. Roy et al. [19] proposed a cross-attention mechanism that utilizes LiDAR data as queries while extracting keys and values from the HS image, enabling transformer-based multimodal feature extraction.

It can be observed that in multimodal LULC classification tasks, the majority of existing works focus on extracting unitary features (either CNN-based or transformer-based) from multimodal data, and these methods typically employ CNNs for low-level feature extraction and transformers for high-level context aggregation. An intuitive idea is that fusing features extracted from different architectures (CNN or transformer) simultaneously can preserve more effective information. More importantly, features derived from diverse architectural approaches applied to multimodal data sources should inherently undergo information interaction and fusion. To clarify these concepts, we naturally propose two distinct levels of information fusion (as shown in Table I): feature level and image level. *Feature-level* fusion refers to integrating multimodal features extracted from a single data source, while *image-level* fusion involves combining unimodal features across multimodal data sources. From this perspective, most current multimodal methods are constrained to either the image or feature level, and do not consider the multimodality of both images and features.

TABLE I
TAXONOMY OF INFORMATION FUSION ARCHITECTURES

Feature Architecture	Image Source	Modality Type	Fusion level
Single-feature	Single-source	Uni-modal	-
Multi-feature	Single-source	Multi-modal	Feature level
Single-feature	Multi-source	Multi-modal	Image level
Multi-feature	Multi-source	Multi-modal	Ours

Building upon these insights, we propose a new and efficient multilevel multimodal fusion strategy, namely, multilevel multimodal CNN–transformer (MCT) fusion network, to address the aforementioned challenges in LULC classification tasks, which uses a multifeature extractor (MFE) to fully leverage the superior spatial information capture capability of CNNs, as well as the powerful global feature representation capability of transformers, and utilizes a cross-architecture fusion (CAF) module to simultaneously fuse multimodal information at both the image and feature levels.

More specifically, our key contributions in this article are summarized as follows.

1) We propose a novel multilevel MCT fusion framework for LULC classification, which integrates HSIs with other multimodal RS data, including LiDAR, SAR, and DSM. The proposed network achieves superior performance through a well-designed fusion network while maintaining a streamlined and generalized architecture.
2) Two novel modules are developed within the MCT fusion network, i.e., MFE and CAF. The MFE extracts multimodal information from diverse images, while the CAF module fuses this information at both feature and image levels. Together, they form a straightforward yet highly effective architecture for multimodal information fusion.
3) We establish a formal taxonomy categorizing multimodal fusion methods into feature-level and image-level paradigms (Table I), and propose a novel architecture that simultaneously bridges both levels, which is a capability absent in prior works. This multilevel fusion framework fundamentally advances multimodal representation learning while offering interpretable design principles.
4) Extensive ablation studies and comparative experiments are conducted on three public HS datasets, i.e., Houston, Berlin, and Augsburg. The results demonstrate the effectiveness and superiority of our proposed approach both qualitatively and quantitatively. We further empirically validate that classification tokens are dispensable when the CAF is employed to fuse multimodal high-level semantics at both image and feature levels.

The remainder of this article is organized as follows. Related work regarding LULC classification of multimodal RS data is introduced in Section II. Section III describes the proposed MCT method, which includes an MFE and a cross-architecture

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information fusion module. The experimental setting and LULC classification results are detailed in Section IV, with analysis based on the ablation study and contrast experiments performed on three benchmark multimodal RS datasets [20]. Section V presents broad conclusions and future research directions.

II. Related Work

A. Traditional Methods for LULC Classification

In LULC classification, traditional classifiers have been widely employed, even with limited training samples. As we described in Section I, these methods usually comprise two stages: feature extraction and classification with traditional classifiers. Specifically, support vector machines (SVMs) with nonlinear kernels have been extensively applied when training samples are limited [8]. Additionally, extreme learning machines (ELMs) have been widely used for feature extraction from imbalanced training datasets. In [9], an ELM was employed to classify HSIs by utilizing local binary patterns (LBPs) for classification. The study revealed that compared with SVMs, ELMs achieved superior classification performance. Another commonly employed classifier is the random forest (RF), which uses an ensemble of decision trees for classification and can effectively capture complex nonlinear relationships in the data [21]. However, traditional methods encounter a performance bottleneck and cannot adaptively capture the intrinsic features as the complexity of the multimodal RS data increases, primarily due to their limited capacity for data fitting and representation.

B. CNN-Based Methods for LULC Classification

Recently, CNNs have shown great potential in multimodal data classification because of their powerful feature representation ability [22], [23]. In contrast to conventional approaches that rely on handcrafted features derived from expert domain knowledge, CNN-based models circumvent this requirement by performing data-driven feature extraction in an end-to-end trainable fashion. Hong et al. [24] proposed the first comprehensive multimodal network framework, which investigates various fusion strategies for heterogeneous RS data. Inspired by the brain-like splitting autoencoder [25], Wu et al. [16] developed a multimodal CNN augmented with cross-channel reconstruction capabilities, aiming to achieve more precise joint classification of multimodal RS data. Due to the significant domain differences among multiple heterogeneous RS datasets, attention mechanisms have been introduced to facilitate more effective multimodal classification by dynamically prioritizing the most salient cross-modal features. In [26], SA was introduced to create an attention map for the HS modality, and a cross-attention module was subsequently designed to exploit the LiDAR-derived attention map, emphasizing the spatial characteristics of HSIs. Wang et al. [27] introduced a multiattentive hierarchical fusion network, which utilized modality attention and SA modules to generate integrated modality features and specific features, respectively. These two feature representations were, then, combined to establish a comprehensive classification framework. In brief, CNNs are adept at capturing spatial patterns and relationships within data through their localized convolutional filters and hierarchical architecture, making them well-suited for tasks involving spatial data modalities such as LULC classification.

C. Transformer-Based Methods for LULC Classification

While CNNs have demonstrated excellent performance in LULC classification, they are limited in their ability to handle the long-sequence attributes of spectral signatures. This is primarily attributed to the inherent network architecture of CNNs, which may not be capable of capturing the intricate patterns present in spectral data. This constraint highlights the need for alternative models that can effectively capture long-range dependencies in sequential data. Transformers have emerged as promising solutions in this regard, leveraging SA mechanisms to efficiently process and analyze sequential data [14], [28], [29], [30]. In the context of RS, transformer-based models offer the potential to overcome the challenges posed by long-range dependencies in spectral signatures, thereby improving the accuracy and effectiveness of LULC classification tasks. Among the recent pioneering contributions in this field, SpectralFormer [17] stands out as the first ViT-based framework to approach HSI analysis from a sequential viewpoint within the spectral domain. Xue et al. [18] introduced a hierarchical ViT architecture to integrate HS and LiDAR data. Leveraging its robust capacity for modeling long-range dependencies, the model demonstrates strong generalizability in capturing discriminative features across modalities. Yao et al. [6] proposed a multimodal DL framework called ExViT by extending conventional ViT with separable convolution modules, offering an economical solution that leverages both spatial and modality-specific channel information. Considering that ViTs and other similar transformers use an external CLS token, which is randomly initialized and often fails to generalize well, Roy et al. [7] proposed a multihead cross-patch attention module to achieve better generalization, in which the CLS token was generated from a corresponding patch taken from other modalities (such as LiDAR, SAR, and DSM). To fuse multimodal heterogeneous information and enhance the joint classification performance, a multimodal transformer network (MTNet) that uses multiple transformers to capture individual and shared features of multimodal RS data was presented by Zhang et al. [31]. Although transformers demonstrate proficiency in capturing the information embedded in spectral signatures, they may fall short in adequately characterizing local semantic elements and leveraging spatial information effectively.

D. CNN–Transformer-Based Methods for LULC Classification

The limitations of standalone CNN or transformer approaches have spurred the development of hybrid architectures in the field of computer vision. These methods typically employ CNNs for low-level feature extraction and transformers for high-level context aggregation. Representative works include ExViT's [6] parallel dual-branch design with cross-attention fusion, which utilized two separable convolution

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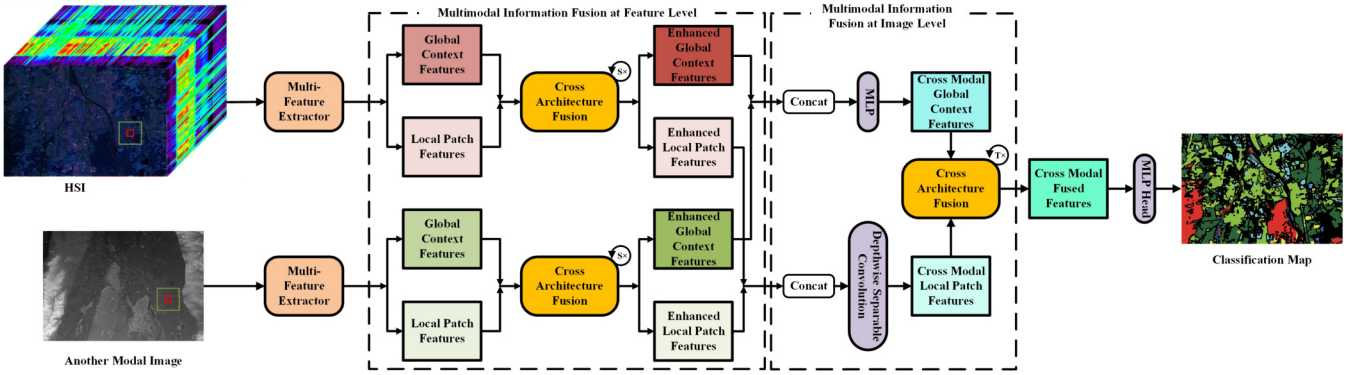


Fig. 1. Overview of the proposed MCT fusion network for LULC classification. The input consists of an HSI and another modal image, where an HSI and DSM image from the Augsburg dataset are utilized for demonstration purposes. Two MFEs are employed to extract multimodal information at the feature level from two inputs. CAF is, then, applied to integrate multimodal information at the feature level. The modality-consistent information obtained from heterogeneous images is subsequently concatenated to obtain multimodal information at the image level. Finally, CAF is performed again at the image level to obtain the real cross-modality fused features.

stems to extract contextual information from multimodal data sources, and a cross-modality attention to promote information exchange across heterogeneous modalities. Ji et al. [32] proposed a CNN–transformer cascade framework to leverage transformer and CNN complementary strengths for long-range and local feature extraction for LULC classification. Different from the dual-branch of ExViT, Lu et al. [33] proposed a dual-branch encoder, where Swin Transformer [34] captures global features while CNN extracts local features. Several CNN–transformer hybrid approaches from semantic segmentation tasks also merit attention. In [35], CNNs serve as feature extractors to derive shallow features from multisource data, followed by transformers for deep feature extraction and fusion. Similarly, Wu et al. [36] employ CNNs at the encoder stage to extract multilevel features from high-resolution RS images, while designing a lightweight transformer module as the decoder for information integration. As listed in Table I, existing multimodal fusion approaches can be categorized into two paradigms: feature-level and image-level fusion. To the best of authors' knowledge, current multimodal methods in both LULC classification and semantic segmentation tasks have not yet achieved genuine multimodality simultaneously at both feature and image levels. In sharp contrast to these existing methods, the proposed MCT method leverages diverse feature extraction paradigms (CNN and transformer) to derive multimodal features from multimodal imagery, while simultaneously accomplishing both feature-level and image-level information fusion.

III. Methodology

A. Preprocessing of HSI and Other Modal Data

Let us denote an HSI as $X_h \in \mathbb{R}^{C_1 \times H \times W}$ and a corresponding other modal image (OMI) as $X_o \in \mathbb{R}^{C_2 \times H \times W}$, where H and W represent the height and width of the image, respectively, and C_1 and C_2 refer to the number of spectral bands of the HSI and the OMI (such as LiDAR, SAR, and DSM), respectively. LULC classification aims to assign each pixel into one of the C land cover classes, represented by the set $C = \{y_1, y_2, \ldots, y_c\}$. A single pixel of the HSI can be defined as $m_{i,j} \in X_h$, where $i = 1, \ldots, H$ and $j = 1, \ldots, W$ with $m_{i,j} = \{m_{i,j,1}, \ldots, m_{i,j,C_1}\} \in \mathbb{R}^{C_1}$ containing C_1 spectral bands. Similarly, a single pixel of the OMI can be defined as $n_{i,j} \in X_o$ with $n_{i,j} = \{n_{i,j,1}, \ldots, n_{i,j,C_2}\} \in \mathbb{R}^{C_2}$ containing C_2 spectral bands. Patch extraction is a conventional preprocessing operation for HSIs, where a spectral–spatial cube $x^h_{i,j} \in \mathbb{R}^{C_1 \times p \times p}$ is obtained from the normalized HSI data X_h and a corresponding patch $x^o_{i,j} \in \mathbb{R}^{C_2 \times p \times p}$ is obtained from the other normalized modal image X_o, with neighboring regions of size $(p \times p)$ and centered at the pixel (i, j).

Leveraging the complementary spectral and spatial information in a joint manner allows us to enhance the discriminative ability of the proposed method. To this end, our approach takes the extracted image patches as input to the network architecture. Finally, the training and test samples in each class for X_h and X_o can be denoted as follows:

$$D^{\text{train}} = \left\{\left(x^h_{(i)}, x^o_{(i)}\right), y^{(i)} | i = 1, \ldots M\right\}$$
$$D^{\text{test}} = \left\{\left(x^h_{(i)}, x^o_{(i)}\right), y^{(i)} | i = 1, \ldots N\right\} \quad (1)$$

where $x^h_{(i)} \in \mathbb{R}^{C_1 \times p \times p}$ and $x^o_{(i)} \in \mathbb{R}^{C_2 \times p \times p}$ are chosen from the HSI and the OMI, respectively; M and N represent the number of training and test samples, respectively; and $y^{(i)}$ is the actual class for the ith land cover pixel.

B. Overview of the Multilevel MCT Fusion Architecture

To address the challenge mentioned in Section I, which is that most current multimodal methods cannot integrate multimodal information at both the image and feature levels, we introduce an MCT fusion network, where multimodal information at both the image and feature levels is thoroughly explored and fused. The overall network architecture is shown in Fig. 1, in which only the HSI and DSM image from the Augsburg dataset are considered; however, the concept can be extended to other multimodal datasets as well. The results for different types of multimodal data (i.e., LiDAR, SAR, and DSM) are reported in Section IV-D.

Formally, MFE is employed to extract multimodal information at the feature level from a single image, and the resulting global context features and local patch features, namely, transformer features and CNN features, which are

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extracted under two different feature extraction paradigms by MFE, belong to multimodal information at the *feature level*. To provide a more rigorous description, we express the process as follows:

$$F_{\text{global}}, F_{\text{local}} = \text{MFE}(\text{HSI})$$
$$T_{\text{global}}, T_{\text{local}} = \text{MFE}(\text{OMI}) \tag{2}$$

where F_{global} and F_{local} represent the global context features and local patch features extracted from the HSI, respectively, T_{global} and T_{local} represent those extracted from the OMI, and MFE$(\cdot)$ represents the operations of the MFE module. Specifically, F_{global} and F_{local} are the multimodal information at the feature level, represented as T_{global} and T_{local}, respectively.

It is generally recognized that CNN features capture spatial information and local patterns within images well because of their hierarchical convolutional operations, whereas transformer features are proficient at capturing long-range dependencies and global contextual information across the input sequence because of their SA mechanism. Accordingly, we enter the resulting global context features and local patch features into the CAF module to fuse the multimodal information at the feature level, aiming to obtain global context features enriched with spatial information and local patch features enriched with global contextual information. The process can be formally described as follows:

$$F^{\text{fl}}_{\text{global}}, F^{\text{fl}}_{\text{local}} = \text{CAF}^{(S)}\left(F_{\text{global}}, F_{\text{local}}\right)$$
$$T^{\text{fl}}_{\text{global}}, T^{\text{fl}}_{\text{local}} = \text{CAF}^{(S)}\left(T_{\text{global}}, T_{\text{local}}\right) \tag{3}$$

where $F^{\text{fl}}_{\text{global}}$ and $F^{\text{fl}}_{\text{local}}$ refer to the fused features at the feature level originating from the HSI, while $T^{\text{fl}}_{\text{global}}$ and $T^{\text{fl}}_{\text{local}}$ represent the fused features at the feature level originating from the OMI, and CAF$^{(S)}(\cdot)$ symbolizes the operations of the CAF module, which are applied S times iteratively.

The MCT connects the modality-consistent information from two heterogeneous images at the image level. Specifically, the specific features obtained from two heterogeneous images, such as global context features or local patch features, belong to different modalities at the *image level*. We concatenate the homogeneous features and perform simple information fusion based on their characteristics, where global context features are fused using multilayer perceptron (MLP) and local patch features are fused using depthwise separable convolution (DSC), resulting in cross-modal global context features and cross-modal local patch features. Generalizing the above description, we have the following:

$$I_{\text{global}} = \text{MLP}\left(\text{Concat}\left(F^{\text{fl}}_{\text{global}}, T^{\text{fl}}_{\text{global}}\right)\right)$$
$$I_{\text{local}} = \text{DSC}\left(\text{Concat}\left(F^{\text{fl}}_{\text{local}}, T^{\text{fl}}_{\text{local}}\right)\right) \tag{4}$$

where I_{global} and I_{local} represent the cross-modal global context features and cross-modal local patch features, respectively.

Finally, CAF is performed again at the image level to obtain the real cross-modality fused feature, and then, the resulting feature is passed through the MLP, ultimately producing the final classification results using the following equation:

$$I = \text{CAF}^{(T)}\left(I_{\text{global}}, I_{\text{local}}\right)$$
$$F_{\text{out}} = \text{MLP}(I) \tag{5}$$

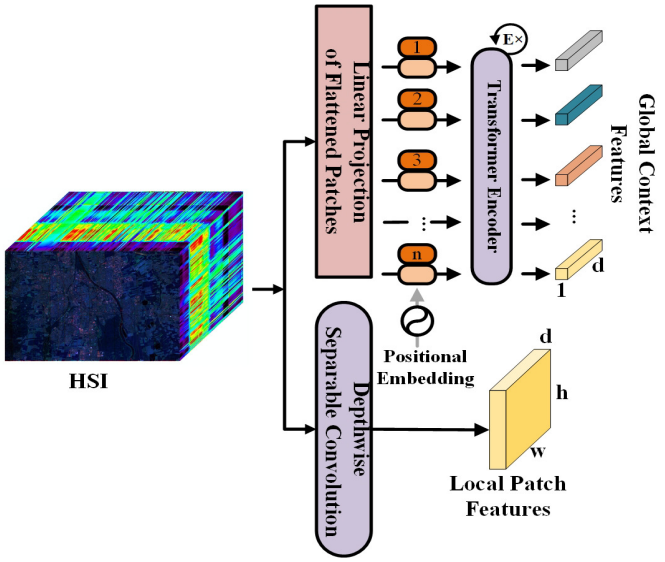


Fig. 2. Detailed design of the proposed MFE, which bidirectionally extracts multimodal features via various feature extraction paradigms. The figure illustrates the input with the HSI, while OMIs can also serve as inputs.

where I denotes the real cross-modality fused feature, CAF$^{(T)}(\cdot)$ represents the operations of the CAF module, which are applied T times iteratively, and F_{out} represents the final number of output vectors and corresponds to the number of classes of specific datasets. After considering the potential semantic discrepancy between transformer attributes and CNN characteristics and mutually integrating spectral sequence attributes with spatial features, the MCT achieves the integration of multimodal information at both the image and feature levels through two distinct feature fusion processes. The key components of the proposed MCT fusion network are discussed step by step as follows.

C. Multifeature Extractor

Both CNNs and transformers demonstrate effectiveness in performing LULC classification tasks because of their strengths in contextual modeling and spectral information utilization. CNNs utilize automatic contextual modeling to make strong inferences and benefit from the large number of spectral bands in HSIs to control the depth of output feature maps. Moreover, transformers employ SA mechanisms to capture long-range dependencies and model contextual information across features, leveraging the rich spectral information in HSIs to increase the discriminative power of the classification model. Hence, as shown in Fig. 2, to obtain multimodal information at the feature level, the proposed MFE employs CNNs and transformers in a parallel manner to perform feature extraction. To facilitate description, in this section, we uniformly denote the patch input as follows:

$$X = [x_1, \ldots, x_i, \ldots, x_c] \in \mathbb{R}^{h \times w \times c} \tag{6}$$

where $x_i \in \mathbb{R}^{h \times w}$ denotes the ith spectral band, c refers to the number of channels of the patch, and h and w are the height and width of the patch, respectively.

To extract robust and discriminative multimodal features from raw images, we explore an alternative network with two

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branches, one using the widely used DSC [37] and the other using a regular ViT encoder [38]. Fig. 2 shows a graphical representation of the DSC and ViT encoder. The following provides the details of each branch.

1) CNN Branch: The adopted DSC is constructed by stacking L blocks. Each block comprises a depthwise (DW) convolution layer, which performs independent 3×3 spatial convolutions on each channel of the previous layer's output. This is followed by a pointwise (PW) convolution layer that applies 1×1 convolutions to integrate and map the channels produced by the DW convolution into a new channel space. Additionally, a nonlinear activation layer using the Gaussian error linear unit function [39] is incorporated, which is consistent with the ViT encoders employed. The DSC design enables efficient extraction of spatial features while reducing computational complexity. By combining the abovementioned operations into a whole learnable function $g(\cdot)$ and obeying conventional DSC [37], we, then, have the following local patch features $F_{\text{local}} \in \mathbb{R}^{h \times w \times d}$ for the patch input:

$$F_{\text{local}} = g(X). \tag{7}$$

2) Transformer Branch: Motivated by SpectralFormer [17], we introduce a novel approach to learn groupwise spectral embeddings for spectral images, which departs from the conventional bandwidth representation paradigm. As described in Section III-A, we take the extracted patches as input to the network, where the patches from the HSI and OMI are extracted from different modality images but undergo the same processing within the network modules. The patch input X can be reshaped along the spatial dimensions as follows:

$$\hat{X} = \left[\vec{x_1}, \ldots, \vec{x_i}, \ldots, \vec{x_c}\right] \in \mathbb{R}^{n \times c} \tag{8}$$

where $\vec{x_i} \in \mathbb{R}^{n \times 1}$ with $(n = h \times w)$ denotes the unfolded, spatially flattened patch for the ith spectral band. This input formulation enables the network to maintain the spatial information from the original patch representations while incorporating a groupwise spectral context during the feature learning stage. The flattened patch is, then, processed following the regular tokenization rule, namely, the linear projection, resulting in a series of 1-D spectral embedding tokens $F' \in \mathbb{R}^{n \times d}$ that are ready for the subsequent transformer encoder. Using a learnable function $h(\cdot)$ to represent the linear projection, we can obtain the following tokenization form for the patch:

$$F' = h\left(\hat{X}\right). \tag{9}$$

The spectral embedding tokens are, then, sent into the transformer encoder for subsequent mapping and feature extraction. Notably, we employ the transformer encoder as a multimodal feature extractor to extract spectral features distinct from spatial features rather than for final classification features. Moreover, we claim that the classification token is not indispensable during feature extraction. Therefore, only the positional embeddings are incorporated into the spectral feature embeddings, while the classification token is omitted. By incorporating positional embeddings, our model not only encodes the position of each token and retains the sequential information of the spectral embedding tokens, but also leverages the advantages of multihead attention (MHA) and captures cross-band dependencies and contextual information. To achieve this, we add the positional embeddings, denoted P, into the obtained spectral embedding tokens as follows:

$$F_0 = F' + P \tag{10}$$

where $P = [p_1, \ldots, p_c]$ represents the positional embeddings, which is an $n \times d$ random matrix. This matrix denotes the position of each token during model training. The feature embedding F_0 is, then, sent to a transformer encoder, which comprises alternating MHA and MLP layers. This transformer encoder is responsible for encoding the spectral sequence attributes and can be represented as follows:

$$\begin{aligned} F'_e &= \text{MHA}\left(\text{LN}\left(F_{e-1}\right)\right) + F_{e-1} \\ F_e &= \text{MLP}\left(\text{LN}\left(F'_e\right)\right) + F'_e \end{aligned} \tag{11}$$

where $\text{LN}(\cdot)$ represents the layer normalization operation. The spectral feature embedding F_e represents the output of the eth transformer encoder, where e denotes the depth of the encoder, which ranges from 1 to E, and E denotes the number of spectral transformer encoders employed in the model. The MHA and MLP operations are alternated E times, producing the spectral feature embedding $F_E \in \mathbb{R}^{n \times d}$ after processing in the final encoder stage. Hereafter, F_E will be referred to as F_{global}.

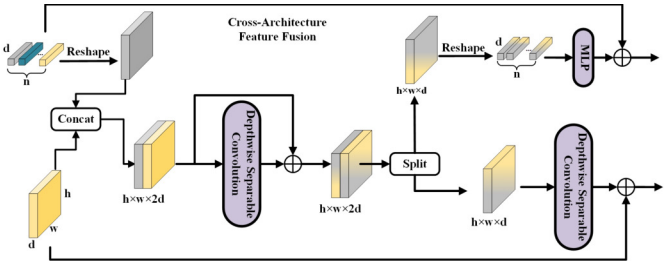


Fig. 3. Detailed design of the proposed CAF. The concatenated features from the two branches are fused using DSCs. The enriched information is, then, bidirectionally transferred back to the original branches.

D. Cross-Architecture Fusion

The intermediate features extracted under two different feature extraction paradigms are bidirectionally connected to fuse the multimodal information at the feature level. Fig. 3 depicts the proposed CAF. The CAF module is designed based on the residual learning principle, where the module learns to generate additive refinements to the original features rather than replacing them entirely. This approach preserves the inherent characteristics of each branch while enabling effective cross-modal information exchange. Specifically, with intermediate features F_{local} and F_{global} from the CNN branch and transformer branch as inputs, respectively, we first concatenate the two features, which results in an intermediate concatenated feature M_{concat}, and then use DSC to initially fuse the features in a residual manner, obtaining the fused feature $M \in \mathbb{R}^{h \times w \times 2d}$. After processing the concatenated features through DSCs, the resulting fused representation M contains enriched cross-architecture information. We subsequently split the fused feature M along its channel dimension, yielding

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two features $M_{\text{local}} \in \mathbb{R}^{h\times w\times d}$ and $M_{\text{global}} \in \mathbb{R}^{h\times w\times d}$ that have undergone information exchange. These features serve as complementary refinements to the original inputs. While these split features do not directly correspond to the original local or global features in terms of channel content, they are specifically designed to enhance their respective branches through additive combination. The M_{local} provides refinements for the local branch, while the M_{global} enhances the global branch, following the residual learning paradigm where the CAF module generates corrective information rather than complete feature replacements, and the above process can be summarized as follows:

$$\begin{aligned} M_{\text{concat}} &= \text{Concat}\left(F_{\text{local}}, \text{Reshape}\left(F_{\text{global}}\right)\right) \\ M &= \text{DSC}\left(M_{\text{concat}}\right) + M_{\text{concat}} \\ M_{\text{local}}, M_{\text{global}} &= \text{Split}\left(M\right) \end{aligned} \tag{12}$$

where DSC(·) denotes the depthwise separable convolution, and Reshape(·) and Concat(·) represent the image-like rearrangement and concatenation, respectively. The Split(·) function divides the fused feature M into two separate features M_{local} and M_{global} along the channel dimension. Before M_{local} and M_{global} are fed back into their respective branches, they are processed through convolutional blocks and MLP blocks, respectively, and then, each feature is individually added to the original input features F_{local} and F_{global}. The procedure is formulated as follows:

$$\begin{aligned} F'_{\text{local}} &= \text{DSC}\left(M_{\text{local}}\right) + F_{\text{local}} \\ F'_{\text{global}} &= \text{MLP}\left(\text{Reshape}\left(M_{\text{global}}\right)\right) + F_{\text{global}} \end{aligned} \tag{13}$$

where F'_{local} and F'_{global} are the fused features enhanced with information from other modalities. This additive combination in (13) implements the residual learning mechanism, allowing the network to learn identity mappings when cross-modal fusion is not beneficial, while enabling effective feature enhancement when cross-modal interactions provide valuable information.

As illustrated in Fig. 1, it should be emphasized that the CAF is repeated S times during feature-level fusion to achieve more thorough information integration, ultimately resulting in feature-level fused features $F^{\text{fl}}_{\text{global}}$ and $F^{\text{fl}}_{\text{local}}$. Different from the feature-level fusion process, the image-level fusion process repeats the CAF module T times, and a key distinction of this stage is the incorporation of two 3×3 convolutional layers in the final CAF module. These convolutional layers serve to reduce the channel dimensionality of the intermediate concatenated features M_{concat}. The iterative stacking design of our CAF module follows the established principle of progressive feature refinement commonly employed in hierarchical architectures [34], [40], [41], where repeated application of fusion modules enables more fine-grained integration compared to single-step fusion processes. Additionally, it is essential to consider that while we refer to CNN-derived features as "local" and transformer-derived features as "global" throughout this article, these semantic distinctions evolve progressively through the iterative CAF operations. The terminology reflects the initial architectural characteristics and processing pReferences rather than absolute semantic boundaries. The CNN branch maintains its sensitivity to fine-grained spatial patterns and local structural information, while the transformer branch preserves its capacity for capturing long-range dependencies and global contextual relationships, even as both feature types become increasingly enriched with complementary information through iterative fusion. Through this process, the network ultimately generates deep features that have undergone multimodal information fusion at both feature and image levels, achieving a more robust representation of the input data.

Inspired by ExViT's demonstration that classification tokens become redundant when comprehensive token fusion is achieved [6], our CAF module inherently eliminates the need for dedicated classification tokens. By performing multimodal fusion at both feature and image levels—where all tokens undergo equal-weighted interaction through CAF—the CAF module naturally consolidates discriminative information across all token representations. This approach aligns with ExViT's finding but extends it through simultaneous cross-source and cross-feature fusion. The resulting unified token embeddings inherently encode classification-ready semantics without classification token injection.

IV. Experiments

In this section, we introduce three widely used HSI datasets along with their associated multimodal data sources (LiDAR, SAR, and DSM) and provide detailed information on the implementation process. In addition, extensive experiments were conducted, including ablation and parameter sensitivity studies, to quantitatively and qualitatively assess the performance of the proposed model.

A. Dataset Description

1) Houston2013 HS-LiDAR Dataset: The Houston2013 HS-LiDAR dataset provided by the University of Houston in June 2012 includes HSI and LiDAR-based DSM data covering the University of Houston campus and surrounding areas. The dataset features a spatial resolution of 2.5 m and consists of 349×1905 pixels. The HSI contains 144 spectral bands ranging from 0.38 to 1.05 μm, whereas the coregistered LiDAR data contain only one channel record of height information. The dataset is annotated with pixel-level labels for 15 LULC classes. We adhere to the training and testing set divisions established by the 2013 IEEE Geoscience and RS Society Data Fusion Contest. This division ensures a clear separation between the datasets, as detailed in Table II. Fig. 4 shows the Houston2013 dataset through multiple visualizations. It includes a false-color representation of the HSI, a grayscale image depicting the LiDAR information, and two maps showcasing the spatial distributions of both the training and testing samples.

2) Berlin Dataset: The Berlin dataset captured by the Sentinel-1A platform (launched on April 3, 2014) in August 2009 includes HSI and SAR data covering the same subset describing Berlin city and its neighboring rural area. The scene contains 1723×476 pixels and has a 13.89-m ground sampling distance (GSD), with

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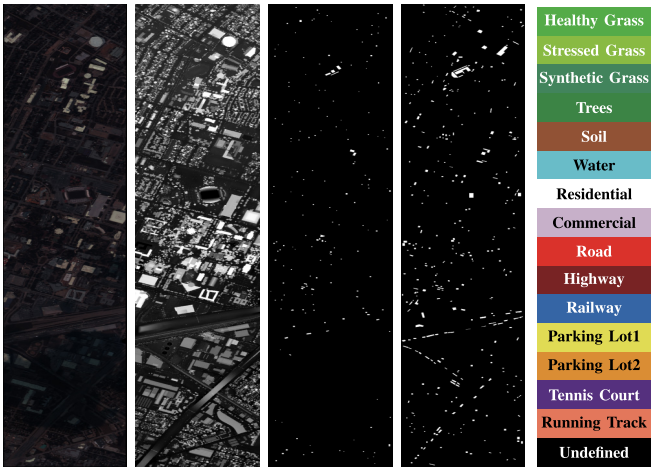


Fig. 4. Houston2013 dataset. (a) False-color image from the HSI. (b) Grayscale image from LiDAR data. (c) Distribution of training images. (d) Distribution of testing images. (e) Legend.

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON2013 DATASET

Class	Class Name	Training	Testing
1	Health Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot1	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
Total	-	2832	12197

the HSI comprising 244 spectral bands spanning the 0.4–2.5-μm wavelength range. The coregistered SAR image includes four channels derived from the covariance matrix of the dual-polarimetric (VH–VV) single-look complex product. We use a ground reference map provided by OpenStreetMap (OSM) and refined it via Google Earth. This map categorizes the area into eight LULC classes, which are detailed in Table III. Fig. 5 shows multiple visualizations of the Berlin dataset. It includes a false-color representation of the HSI, a grayscale image depicting the SAR information, and two maps showcasing the spatial distributions of the training and testing samples.

3) Augsburg Dataset: The Augsburg dataset comprises an HSI, a dual-polarimetric (VH–VV) SAR image, and a DSM image of the city of Augsburg, Germany, which were captured by a HySpex sensor, a Sentinel-1 sensor, and the DLR-3 K system, respectively. The scene contains 332 $\times$ 485 pixels, with the HSI comprising 180 spectral channels that range from 0.4 to 2.5 μm, and all images have a spatial resolution of 30-m GSD. The pixelwise LULC cover references contain seven classes, with the training and testing information summarized

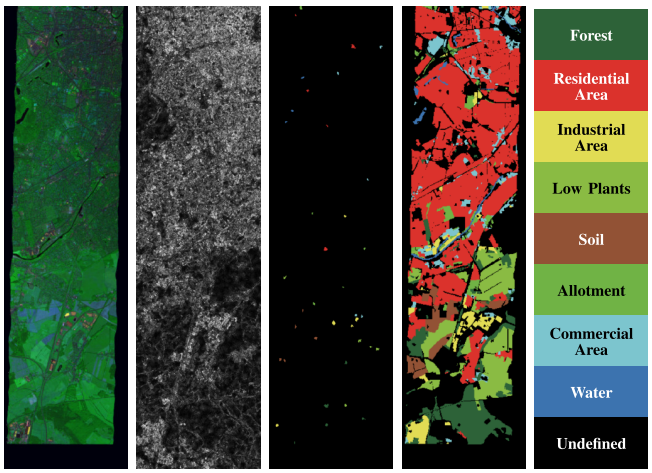


Fig. 5. Berlin dataset. (a) False-color image from the HSI. (b) Grayscale image from SAR data. (c) Distribution of training samples. (d) Distribution of testing samples. (e) Legend.

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE BERLIN DATASET

Class	Class Name	Training	Testing
1	Forest	443	54511
2	Residential Area	423	268219
3	Industrial Area	499	19067
4	Low Plants	376	58906
5	Soil	331	17095
6	Allotment	280	13025
7	Commercial Area	298	24526
8	Water	170	6502
Total	-	2820	461851

in Table IV. Fig. 6 shows multiple visualizations of the Augsburg dataset. It includes a false-color representation of the HSI, a grayscale image depicting the SAR information, a grayscale image describing the DSM data, and a corresponding ground-truth map of this dataset.

B. Experimental Setting

1) Evaluation Metrics: We evaluate the classification performance of each model using three widely used metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). In addition, we visualize the classification maps produced by different models to facilitate a qualitative comparison.

2) Comparison With SOTA Backbone Networks: In the current field of LULC classification, most state-of-the-art (SOTA) methods are based on ViT. Therefore, we select ViT-based methods for the following comparison experiments, such as ViT [38], SpectralFormer [17], ExViT [6], and MTNet [31], along with two traditional classification models, SVM [8] and RF [21], for fair comparison. The experiments were conducted using HSI data combined with OMI sources, including LiDAR, SAR, and DSM. For comparison with the other models being that lack multimodal information processing capabilities, the HSI and OMI data were concatenated bandwise before being

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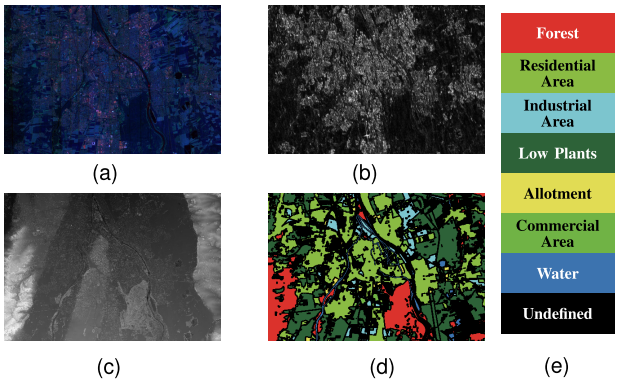


Fig. 6. Augsburg dataset. (a) False-color image from the HSI. (b) Grayscale image from the SAR data. (c) Grayscale image from the DSM data. (d) Ground truth of all samples. (e) Legend.

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE AUGSBURG DATASET

Class	Class Name	Training	Testing
1	Forest	146	13361
2	Residential Area	264	30065
3	Industrial Area	21	3830
4	Low Plants	248	26609
5	Allotment	52	523
6	Commercial Area	7	1638
7	Water	23	1507
Total	-	761	77533

input into the models. The detailed configurations of these models are listed as follows.

1) For the SVM, the sklearn-svc toolbox is chosen for classification tasks. This SVM implementation uses a radial basis function kernel whose key hyperparameters are the regularization parameter, which is empirically set to 100, and the tolerance for the stopping criterion, which is set to $1e^{-7}$.
2) RF was implemented using the sklearn-ensemble, with four key hyperparameters: the number of decision trees, maximum tree depth, minimum samples per leaf node, and maximum features per split. We configured the model with 200 decision trees, a maximum tree depth of 10, a minimum of three samples per leaf node, and ten maximum features per split.
3) For ViT, we followed previous work [38], i.e., only including transformer encoders, while the multimodal data needed to be concatenated along the channel dimension as input.
4) For the Spectralformer, the network employs cross-layer skip connections to transfer memory-like components from shallow to deep layers, adaptively learning to fuse "soft" residuals across different layers. Since it lacks the ability to handle multimodal information, we keep the network architecture unchanged and concatenate the multimodal data along the channel dimension as input to ensure fairness. The remaining hyperparameter settings are consistent with those in the original article.
5) For the ExViT, the architecture handles multimodal RS image patches using parallel branches of a position-shared ViT extended with separable convolution modules, which is an efficient means to utilize both spatial and modality-specific channel information. The remaining hyperparameters were consistent with those specified in the original article.
6) For the MTNet, the model uses multiple transformers to capture the individual and shared features of two modalities. In this model, both multimodal feature extraction and multimodal feature fusion are performed using transformers. Since this model can handle multimodal information, we use the same network and parameter configuration as in the original article for our experiments.

3) Implementation Details: We developed our MCT model using the PyTorch framework on our experimental workstation equipped with 512 GB of RAM and an NVIDIA A10 GPU. To optimize our model parameters, we selected the Adam optimizer with a batch size of 64. The learning rate starts at $5e^{-4}$ and decays by a factor of 0.9 every 20 steps. According to our experiments, the patch size is set to 13 for the HSI and OMI, that is, $p = 13$. The 13×13 pixel image patches are simultaneously processed by both the separable convolution module, which consists of $L = 3$ consecutive blocks, and the transformer encoder within the MFE module. The transformer encoder is configured with the following settings: a hidden layer width of 64, an encoder depth of $E = 6$ (half of the 12 commonly used in natural image classification), four attention heads, and a dropout rate of 0.1. As shown in Fig. 1, two hyperparameters, S and T, are introduced in our framework, which are the number of repetitions for cross-modal information fusion at the feature level and image level, respectively. These parameters are experimentally tuned and fixed at 3 and 4 for optimal performance. Our experiments indicate that training the model for 500 epochs results in a stable and high-performing model when used with the three datasets we tested.

C. Model Analysis

1) Ablation Study: We investigate the performance gain of the proposed MCT in terms of classification accuracy by adding varying numbers of CAFs at different stages and using feature fusion at different levels. For that, we conducted comprehensive ablation experiments on the Houston dataset to verify the effectiveness of our proposed method for LULC classification, as listed in Table V.

The baseline for the ablation studies is similar to that shown in Fig. 1, with the main difference being that it does not include the CAF module. In the MFE, the CNN branch uses DSC [37], whereas the transformer branch consists of the encoder part of the ViT [38].

To evaluate the effectiveness of the CAF module, we design a simple fusion module as a substitute. Specifically, this substitute only applies an MLP for basic fusion on the concatenated features and lacks the branching operations present in the CAF module. We investigated the performance of the network using different numbers of CAF modules at various levels, and the cell located in row "FL0" and column "IL0," denoted as (FL0, IL0), represents the baseline.

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TABLE V

OA, AA, AND κ VALUES ON THE HOUSTON2013 DATASET (IN %) BY CONSIDERING DIFFERENT QUANTITIES AND LEVELS OF THE CAF MODULE. (A) OA. (B) AA. (C) κ COEFFICIENT

(a)

	IL0	IL1	IL2	IL3	IL4
FL0	88.42	89.45	87.85	89.08	89.31
FL1	90.48	88.42	90.07	88.86	89.95
FL2	88.92	89.55	88.01	90.19	89.81
FL3	88.92	89.21	88.60	88.92	**91.37**
FL4	89.83	89.52	88.72	89.77	88.98

(b)

	IL0	IL1	IL2	IL3	IL4
FL0	90.29	90.61	87.90	90.74	90.62
FL1	91.93	90.29	90.66	90.16	90.92
FL2	90.42	90.84	89.38	90.40	90.97
FL3	90.31	90.78	89.81	90.31	**91.89**
FL4	90.86	91.03	90.26	90.93	90.24

(c)

	IL0	IL1	IL2	IL3	IL4
FL0	87.46	88.57	86.84	88.16	88.42
FL1	89.67	87.46	89.22	87.94	89.09
FL2	87.98	88.66	86.99	89.35	88.94
FL3	87.99	88.31	87.67	88.00	**90.64**
FL4	88.96	88.65	87.76	88.90	88.06

The FL{s} indicates adding s CAFs at the Feature Level, and the IL{t} indicates adding t CAFs at the Image Level, higher values indicate better performance.

As shown in Table V, once the CAF module is utilized, the performance surpasses that of the baseline. Furthermore, in terms of the OA, AA, and κ metrics, (FL3, IL4) achieves the best results. This indicates that using three CAF modules for feature fusion at the feature level and four CAF modules for feature fusion at the image level yields the optimal performance. Moreover, the ablation studies indicate that more CAF modules do not necessarily lead to better performance. For instance, (FL4, IL4) performs poorly across all the metrics. Although the experimental results do not show a clear linear pattern, we observe that the performance is generally better when the number of CAF modules at the image level exceeds that at the feature level. Taking the OA metric as an example, multiple configurations in the top-right portion of Table V(a) achieve a performance over 90%, outperforming those in the bottom-left portion. This suggests that information fusion at the image level is more important, while also indicating that the CAF modules at the feature level already achieve effective information fusion, thereby ensuring better results at the image level. Therefore, in the following comparative experiments, we adopt the parameter configuration corresponding to (FL3, IL4).

2) Parameter Sensitivity Analysis: During the training stage, the spatial size of the input patches is a vital parameter that influences the performance of the proposed MCT network [17]. Hence, identifying the optimal parameter size is essential for achieving favorable classification results. As shown in Fig. 7, it is clear that the patch size significantly impacts the performance of the proposed method. As the patch size increases, all three metrics initially improve, reaching their highest values at a patch size of 13. Specifically, OA and kappa increase, peaking at 91.37% and 90.64%, respectively,

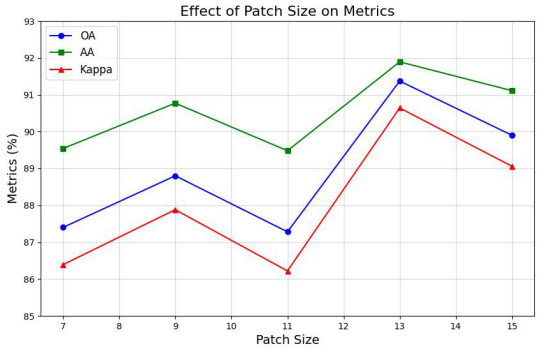


Fig. 7. OA, AA, and kappa coefficient (κ) obtained on the Houston datasets (pixel size: 2.5 m) by the MTNet architecture with different patch sizes.

TABLE VI

EFFICIENCY OF THE REFERENCE NETWORKS AND THE PROPOSED FRAMEWORK ON THE HOUSTON2013 DATASET

Metric	ViT	SpectralFormer	ExViT	MTNet	MCT
Params(M)	0.228	0.237	0.24	0.813	0.8
FLOPs(M)	12.82	22.39	45.21	82.68	134.8
OA(%)	78.68	83.76	89.61	88.73	91.37
AA(%)	81.12	85.28	91.29	89.51	91.89
κ(%)	76.95	82.42	88.75	87.76	90.64

whereas AA achieves its best performance of 91.89% at this patch size. Notably, at a patch size of 11, all three metrics exhibit a temporary decline. This decrease may be attributed to insufficient contextual information being captured by the model due to the intermediate patch size, which fails to balance the tradeoff between local details and global spatial features. Beyond patch size of 13, the metrics begin to decline again, likely due to the inclusion of redundant or irrelevant information in overly large patches. Therefore, to achieve optimal classification performance, a patch size of 13 is selected for subsequent experiments.

3) Model Complexity Analysis: Model complexity serves as another crucial performance indicator. In this section, we provide a quantitative analysis of the model's complexity from both temporal and spatial perspectives. Specifically, we use floating-point operations (FLOPs) to measure time complexity, while the number of parameters evaluates space complexity. Table VI presents the detailed parameter counts and FLOPs for each transformer-based model on the Houston2013 dataset. From this table, we can observe that the proposed MCT network demonstrates significant computational advantages over other architectures. With 0.8M parameters and 134.8M FLOPs, MCT achieves the highest performance across all metrics (OA: 91.37%, AA: 91.89%, and κ: 90.64%) while maintaining remarkable efficiency. Compared to the heaviest model MTNet (0.813M parameters and 82.68M FLOPs), MCT uses nearly the same parameter count but requires 63% more FLOPs, yet delivers substantially better accuracy improvements (+2.64% OA, +2.38% AA, and +2.88% κ). Notably, MCT's FLOPs are higher than counterparts due to the stacked CAF modules, which is a deliberate tradeoff where repeated cross-architecture operations enable progressive refinement of multimodal feature interactions, thus achieving superior multimodal information fusion performance. More

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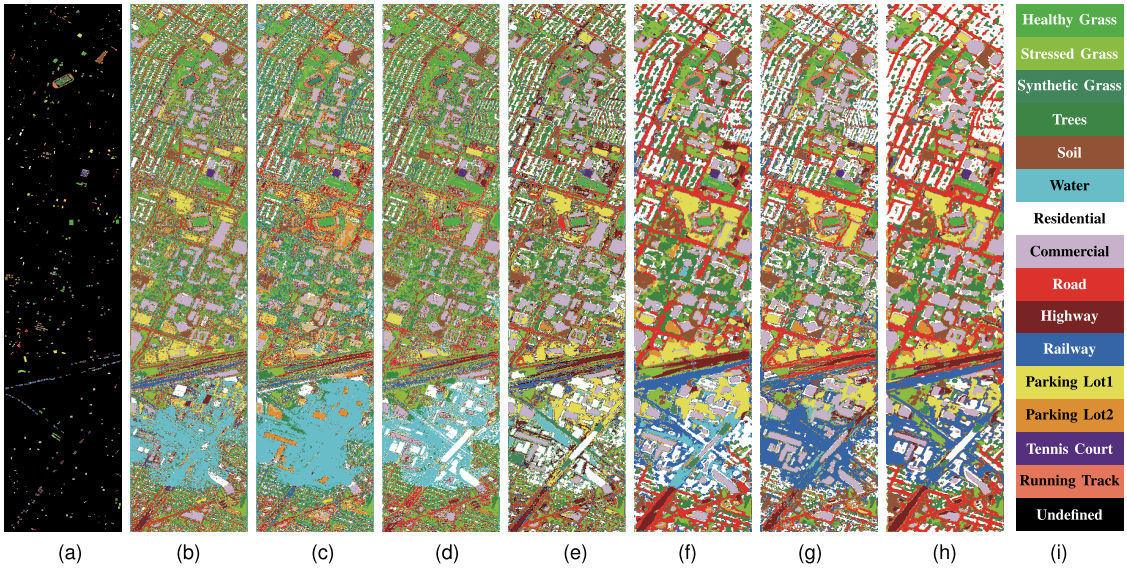


Fig. 8. Visualization of classification maps of the Houston2013 dataset by different models: (a) ground truth; (b) SVM; (c) RF; (d) ViT; (e) SpectralFormer (patchwise); (f) ExViT; (g) MTNet; (h) MCT; and (i) legend. Note: Black regions represent unlabeled areas, consistent with Houston2013 benchmark protocols [20].

TABLE VII

PERFORMANCE OF THE SOTA METHODS AND THE PROPOSED FRAMEWORK IN TERMS OF OA (%), AA (%), AND κ VALUES (%) ON THE HOUSTON2013 DATASET

Class Name	SVM	RF	ViT	SpectralFormer (pixel-wise)	SpectralFormer (patch-wise)	ExViT	MTNet	MCT
Health Grass	82.15	82.53	81.67	**82.81**	82.91	81.96	82.62	82.24
Stressed Grass	81.30	83.55	78.48	90.98	77.54	84.68	84.68	**96.90**
Synthetic Grass	**99.80**	97.43	**99.80**	**99.80**	96.44	98.61	87.92	93.07
Tree	98.48	91.38	**98.67**	92.80	92.99	92.99	91.38	97.06
Soil	98.30	96.31	98.11	98.48	99.62	**100.00**	**100.00**	**100.00**
Water	**99.30**	95.10	95.10	98.60	94.41	**99.30**	97.20	90.21
Residential	80.50	80.22	81.25	83.40	86.01	94.22	94.22	**94.78**
Commercial	58.31	38.08	71.70	74.83	79.58	93.83	**94.49**	87.84
Road	80.93	70.54	77.90	77.43	85.65	88.29	79.70	**91.12**
Highway	48.45	36.87	39.38	57.92	63.03	66.02	**71.53**	65.83
Railway	81.40	73.15	71.63	89.18	72.49	89.66	90.42	**97.15**
Parking Lot1	76.95	51.30	66.09	83.19	84.25	92.32	94.14	**94.91**
Parking Lot2	70.18	54.04	57.89	73.33	65.96	**89.82**	84.91	87.72
Tennis Court	**100.00**	99.60	**100.00**	**100.00**	99.60	**100.00**	94.33	**100.00**
Running Track	98.73	95.77	99.15	98.52	98.73	97.67	95.14	**99.58**
OA	80.89	73.10	78.68	84.73	83.76	89.61	88.73	**91.37**
AA	83.67	76.39	81.12	86.75	85.28	91.29	89.51	**91.89**
$\kappa \times 100$	79.40	71.07	76.95	83.44	82.42	88.75	87.76	**90.64**

importantly, when compared to lightweight alternatives like ViT (0.228M parameters and 12.82M FLOPs), MCT achieves dramatically superior performance (+12.69% OA, +10.77% AA, and +13.69% κ) with a reasonable increase in computational cost, demonstrating an excellent accuracy–efficiency tradeoff. In addition to the optimal MCT setting (FL3, IL4), we also highlight the (FL1, IL0) configuration as a competitive suboptimal choice. With only 0.43M parameters and 68.02M FLOPs, this configuration achieves an OA of 90.48%, which is remarkably close to the optimal configuration (FL3, IL4). This suggests that (FL1, IL0) strikes an excellent balance between efficiency and accuracy, making it particularly suitable for resource-constrained applications.

D. Comparison Results and Analysis

1) Houston2013 HS-LiDAR Dataset: We first present the classification results of different models using HSI and LiDAR data from the Houston 2013 dataset. Table VII provides the quantitative metrics, including OA, AA, κ, and per-class accuracies, while the corresponding classification maps are illustrated in Fig. 8.

In general, traditional methods such as SVM and RF achieve moderate performance, with OA values of 80.89% and 73.10%, respectively. The vanilla ViT model showed comparable performance (OA: 78.68%) to traditional methods. SpectralFormer, with both pixelwise and patchwise variants, demonstrated improved performance, with OA values reaching 84.73% and 83.76%, respectively. ExViT and MTNet further enhanced the performance, achieving OA values of 89.61% and 88.73%, respectively. Notably, our proposed MCT method outperformed all the competitors, achieving the highest OA (91.37%), AA (91.89%), and κ coefficient (90.64%). In particular, MCT showed superior performance in challenging classes such as commercial areas (87.84%) and railways (97.15%),

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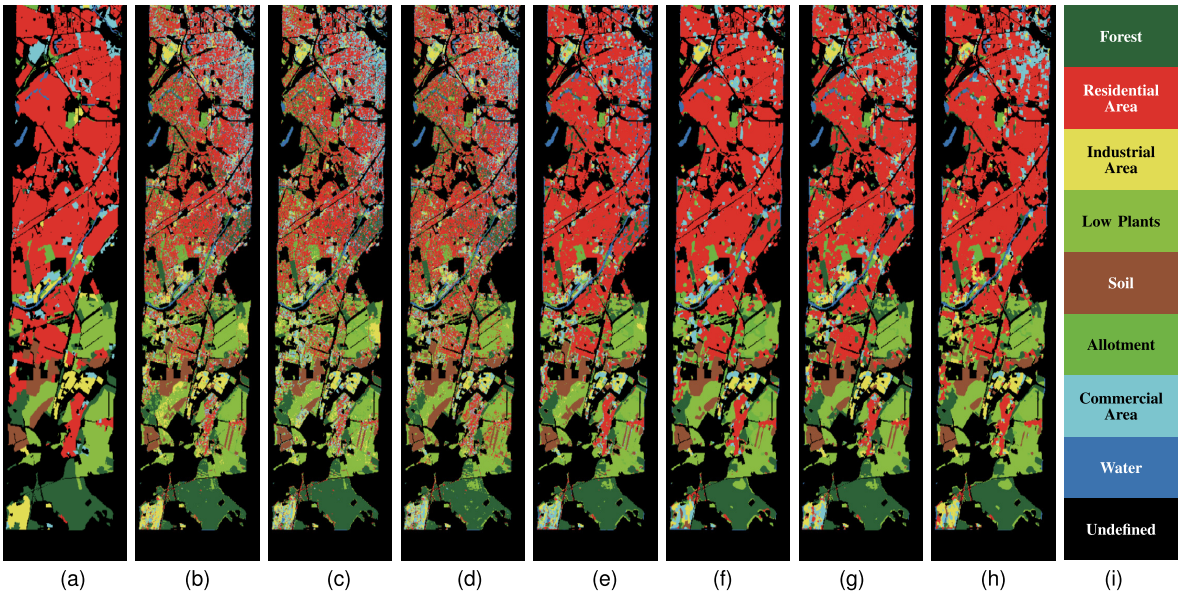


Fig. 9. Visualization of classification maps of the Berlin dataset by different models: (a) ground truth; (b) SVM; (c) RF; (d) ViT; (e) SpectralFormer (patchwise); (f) ExViT; (g) MTNet; (h) MCT; and (i) legend.

TABLE VIII

PERFORMANCE OF THE SOTA METHODS AND THE PROPOSED FRAMEWORK IN TERMS OF OA (%), AA (%), AND κ VALUES (%) ON THE BERLIN DATASET

Class Name	SVM	RF	ViT	SpectralFormer (pixel-wise)	SpectralFormer (patch-wise)	ExViT	MTNet	MCT
Forest	79.54	79.17	77.29	81.34	75.04	66.87	**84.19**	75.4
Residential Area	54.34	51.26	53.75	59.59	75.43	**84.38**	77.16	80.92
Industrial Area	**64.16**	38.90	32.68	44.35	26.62	31.15	49.37	46.83
Low Plants	79.23	68.88	75.48	79.54	76.98	77.06	83.99	**88.91**
Soil	76.34	67.69	86.64	77.96	**85.53**	82.86	80.75	72.38
Allotment	**66.16**	60.54	49.85	54.45	48.01	51.30	62.86	55.27
Commercial Area	30.92	36.11	33.58	33.67	37.27	**38.86**	32.76	27.57
Water	66.79	66.32	68.87	68.09	82.13	67.84	79.81	**80.18**
OA	60.56	56.57	58.68	63.35	71.23	75.54	75.12	**76.00**
AA	63.44	58.61	59.77	62.37	63.37	62.54	**68.86**	65.93
$\kappa \times 100$	47.43	42.72	44.34	49.61	57.78	62.46	63.51	**63.62**

and achieved perfect classification (100%) for certain classes such as the tennis courts and soil. These results demonstrate the effectiveness and robustness of our MCT framework in HSI classification tasks.

The classification maps in Fig. 8 provide a visual comparison of different models applied to the Houston2013 dataset. Traditional methods such as SVM and RF exhibit significant misclassifications, particularly in complex regions with mixed land covers, such as highways and residential areas, which appear noisy and lack clear boundaries. The ViT demonstrates an improvement over these traditional approaches, but it still struggles with fine-grained details in heterogeneous regions. SpectralFormer (patchwise), ExViT, and MTNet further refine the classification, resulting in reduced noise and improved delineation of class boundaries, especially for roads and residential areas. Among all the models, our proposed MCT method achieves the most accurate and visually coherent classification map, with sharper boundaries, reduced noise, and precise identification of challenging classes such as "Road," "Railway," "Residential," and "Parking Lot1." The superior performance of the MCT in capturing multimodal information effectively and fusing it at both the feature and image levels is evident, ensuring accurate and consistent predictions across the dataset.

2) Berlin Dataset: For further evaluation, we tested the methods on the second Berlin dataset using HSI and SAR data. Similarly, the quantitative and visual classification results are shown in Table VIII and Fig. 9, respectively.

The experimental results on the Berlin dataset reveal varying performances across different methods. A notable pattern emerges when comparing patch-based and nonpatch-based approaches: in terms of OA, methods that do not utilize patches as input (SVM: 60.56%, RF: 56.57%, ViT: 58.68%, and SpectralFormer pixelwise: 63.35%) consistently achieve lower OA values than patch-based methods do (SpectralFormer patchwise: 71.23%, ExViT: 75.54%, MTNet: 75.12%, and MCT: 76.00%). Moreover, this same trend is evident in the κ coefficient values. This consistent pattern across both metrics strongly suggests the effectiveness of patch-based processing in capturing spatial–spectral relationships. Among all methods, our proposed MCT demonstrates competitive performance, achieving the highest OA of 76.00% and a κ coefficient of 63.62%. MCT showed particular strengths in challenging categories, achieving superior accuracy in several

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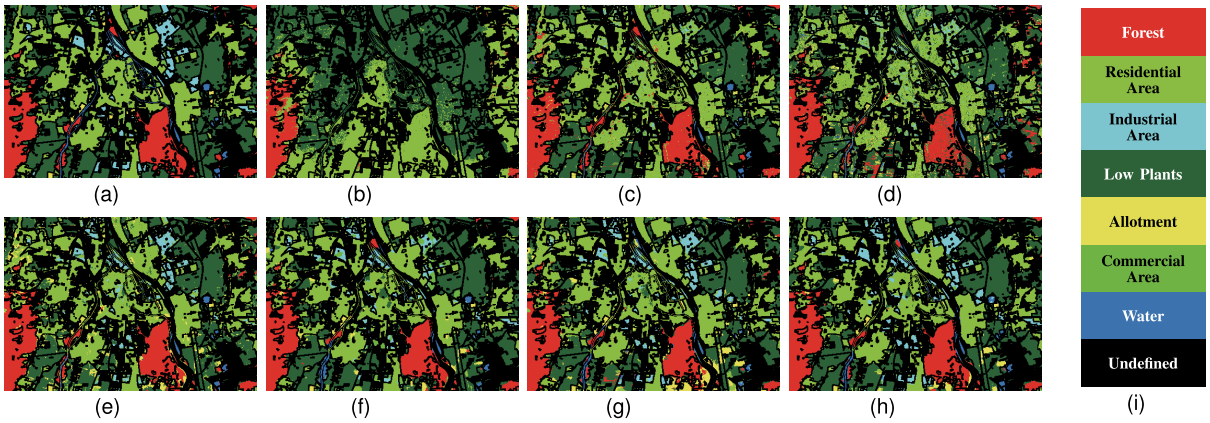


Fig. 10. Visualization of classification maps obtained from HSI and DSM data for the Augsburg dataset by different models: (a) ground truth; (b) SVM; (c) RF; (d) ViT; (e) SpectralFormer (patchwise); (f) ExViT; (g) MTNet; (h) MCT; and (i) legend.

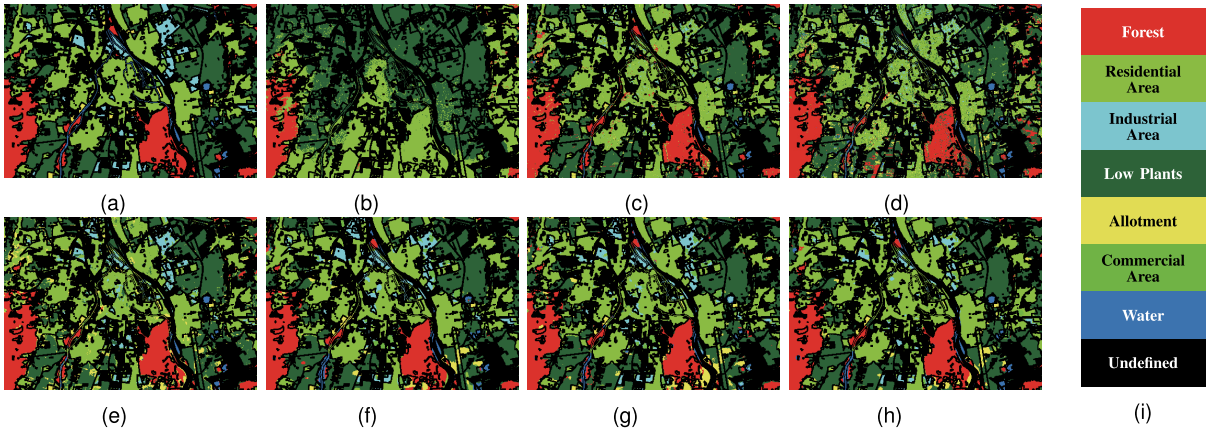


Fig. 11. Visualization of classification maps obtained from HSI and SAR data for the Augsburg dataset by different models: (a) ground truth; (b) SVM; (c) RF; (d) ViT; (e) SpectralFormer (patchwise); (f) ExViT; (g) MTNet; (h) MCT; and (i) legend.

key classes, such as "Low Plants" (88.91%) and "Water" (80.18%). These results demonstrate that the MCT maintains robust performance even on the complex Berlin dataset, which combines HS and SAR data, further validating its effectiveness across different types of RS data fusion tasks.

For the visual comparison of classification maps on the Berlin dataset, as shown in Fig. 9, the proposed MCT stands out for offering the most visually accurate results, closely aligning with the ground truth. These visual improvements align well with the quantitative metrics and further demonstrate the superiority of our MCT framework in handling complex LULC tasks because of its ability to effectively integrate multimodal data.

3) Augsburg Dataset: To demonstrate the generalization ability of our proposed MCT framework in handling multiple data modalities, we evaluated its performance on the Augsburg dataset using three different types of input data: HSI, DSM, and SAR data.

We conducted two sets of experiments using data from three different modalities: HSI with DSM and HSI with SAR. The corresponding classification results and their visualizations are presented in Table IX and Fig. 10 and Table X and Fig. 11. As mentioned in Section IV-B.2, for models lacking multimodal processing capabilities, such as SVM, RF, ViT, and SpectralFormer, the HSI and OMI data were concatenated along the channel dimension before being input into the models. Therefore, the experimental data for these models are reused in both sets of experiments.

A detailed comparison of the experimental results on the Augsburg dataset using different modality combinations (HSI + DSM and HSI + SAR) reveals several interesting patterns that demonstrate the superiority of our MCT method. First, the MCT method achieves the highest overall performance metrics in both scenarios, with OA values of 89.51% (HSI + DSM) and 88.91% (HSI + SAR) and κ coefficients of 84.97% and 84.00%, respectively. This consistently high performance across different modality combinations demonstrates the MCT method's robust ability to effectively fuse and utilize heterogeneous data sources. More specifically, the MCT method shows remarkable stability in handling challenging classes. For example, in both modality combinations, it maintains strong performance in classifying "Forest" (93.20% with DSM and 95.17% with SAR) and "Residential Area" (96.13% with DSM and 97.70% with SAR). This consistency is particularly noteworthy compared with other methods, which often show significant performance variations between the two modality combinations. Another notable

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TABLE IX

PERFORMANCE OF THE SOTA METHODS AND THE PROPOSED FRAMEWORK IN TERMS OF OA (%), AA (%), AND κ VALUES (%) ON THE AUGSBURG (HSI AND DSM) DATASET

Class Name	SVM	RF	ViT	SpectralFormer (pixel-wise)	SpectralFormer (patch-wise)	ExViT	MTNet	MCT
Forest	45.98	**94.21**	88.36	93.98	92.70	92.37	88.53	93.20
Residential Area	39.85	93.52	80.90	90.61	92.75	94.97	**96.53**	96.13
Industrial Area	0.00	15.56	28.98	35.48	60.73	41.36	70.89	**85.09**
Low Plants	59.13	**95.16**	82.81	85.84	83.31	90.28	81.43	89.51
Allotment	0.00	34.80	35.37	42.45	**62.52**	43.79	57.36	44.93
Commercial Area	0.00	12.94	13.98	16.36	6.47	**28.75**	2.32	4.09
Water	0.00	43.66	**47.18**	47.11	38.29	42.00	37.76	44.66
OA	43.67	87.28	77.90	84.09	84.84	87.49	85.31	**89.51**
AA	20.71	55.69	53.94	58.83	62.40	61.93	62.12	**65.37**
$\kappa \times 100$	13.82	81.41	68.49	76.96	78.39	82.05	79.02	**84.97**

TABLE X

PERFORMANCE OF THE SOTA METHODS AND THE PROPOSED FRAMEWORK IN TERMS OF OA (%), AA (%), AND κ VALUES (%) ON THE AUGSBURG (HSI AND SAR) DATASET

Class Name	SVM	RF	ViT	SpectralFormer (pixel-wise)	SpectralFormer (patch-wise)	ExViT	MTNet	MCT
Forest	45.98	94.21	88.36	93.98	92.70	93.43	89.57	**95.17**
Residential Area	39.85	93.52	80.90	90.61	92.75	**97.92**	97.56	97.70
Industrial Area	0.00	15.56	28.98	35.48	60.73	59.97	**62.74**	42.22
Low Plants	59.13	**95.16**	82.81	85.84	83.31	86.17	87.15	90.21
Allotment	0.00	34.80	35.37	42.45	62.52	45.89	43.40	**47.23**
Commercial Area	0.00	12.94	13.98	16.36	6.47	**25.64**	21.92	16.24
Water	0.00	43.66	**47.18**	47.11	38.29	41.34	31.39	47.05
OA	43.67	87.28	77.90	84.09	84.84	88.26	87.64	**88.91**
AA	20.71	55.69	53.94	58.83	62.40	**64.34**	61.96	62.26
$\kappa \times 100$	13.82	81.41	68.49	76.96	78.39	83.23	82.35	**84.00**

TABLE XI

STATISTICAL SIGNIFICANCE (Z-SCORE) ANALYSIS OF OA BETWEEN MCT AND OTHER METHODS

Method A	Method B	Houston2013(n=12197)				Berlin(n=461851)				Augsburg(n=77533) (HSI and DSM)				Augsburg(n=77533) (HSI and SAR)			
		OA(A)	OA(B)	Z-score	Significance	OA(A)	OA(B)	Z-score	Significance	OA(A)	OA(B)	Z-score	Significance	OA(A)	OA(B)	Z-score	Significance
MCT	SVM	91.37%	80.89%	23.96	Yes($p < 0.01$)	76.00%	60.56%	161.67	Yes($p < 0.01$)	89.51%	43.67%	218.93	Yes($p < 0.01$)	88.91%	43.67%	214.59	Yes($p < 0.01$)
MCT	RF	91.37%	73.10%	38.44	Yes($p < 0.01$)	76.00%	56.57%	201.81	Yes($p < 0.01$)	89.51%	87.28%	13.72	Yes($p < 0.01$)	88.91%	87.28%	9.91	Yes($p < 0.01$)
MCT	ViT	91.37%	78.68%	28.22	Yes($p < 0.01$)	76.00%	58.68%	180.58	Yes($p < 0.01$)	89.51%	77.90%	62.67	Yes($p < 0.01$)	88.91%	77.90%	58.96	Yes($p < 0.01$)
MCT	SpectralFormer	91.37%	83.76%	18.13	Yes($p < 0.01$)	76.00%	71.23%	52.09	Yes($p < 0.01$)	89.51%	84.84%	27.57	Yes($p < 0.01$)	88.91%	84.84%	23.77	Yes($p < 0.01$)
MCT	ExViT	91.37%	89.61%	4.68	Yes($p < 0.01$)	76.00%	75.54%	5.16	Yes($p < 0.01$)	89.51%	87.49%	12.47	Yes($p < 0.01$)	88.91%	88.26%	4.02	Yes($p < 0.01$)
MCT	MTNet	91.37%	88.73%	6.89	Yes($p < 0.01$)	76.00%	75.12%	9.84	Yes($p < 0.01$)	89.51%	85.31%	24.98	Yes($p < 0.01$)	88.91%	87.64%	7.77	Yes($p < 0.01$)

observation is that the MCT method consistently outperforms both traditional methods (SVM and RF) and other transformer-based approaches (ViT, SpectralFormer variants, ExViT, and MTNet) across most metrics in both modality combinations. This consistently superior performance across different data fusion scenarios strongly validates the effectiveness of our proposed MCT framework in multimodal LULC classification tasks.

The visualization results in Figs. 10 and 11 further validate the aforementioned analysis. These figures clearly demonstrate the superior performance of the MCT framework in accurately identifying complex and diverse land cover classes compared with other methods.

4) Statistical Significance Analysis: To further validate the performance differences between our proposed MCT framework and other baseline methods, we conducted statistical significance tests based on Z-scores using the OA metric across three datasets. The results are summarized in Table XI. Each classification outcome was treated as an independent Bernoulli trial (correct versus incorrect) for the purpose of Z-score calculation, following standard practice in classification performance evaluation. Across all datasets, the OA improvements of MCT over the baselines are statistically significant with $p < 0.01$, confirming that the observed accuracy gains are not due to random variation. Even in comparisons with strong baselines such as ExViT and MTNet, MCT achieves significant improvements (e.g., $Z = 4.68$ and $Z = 6.89$ on Houston2013), further supporting the robustness of our design. These results demonstrate that MCT consistently delivers not only higher OA but also statistically reliable performance gains.

V. CONCLUSION

In this work, we propose a multilevel MCT fusion network to overcome the critical limitations of conventional single-level fusion approaches in LULC classification. Unlike existing methods that perform shallow fusion at isolated stages, our framework establishes a systematic multilevel fusion paradigm for simultaneous image-level and feature-level fusion. The proposed MCT network effectively integrates multimodal

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information at both the image and feature levels through the MFE and CAF modules and the whole multilevel MCT fusion architecture. By leveraging the spatial information extraction strength of the CNN and the global representation capability of the transformer, the MCT network achieves superior performance in LULC classification tasks.

Comprehensive experiments conducted on three benchmark multimodal datasets, "Houston2013, Berlin, and Augsburg," demonstrated the effectiveness and robustness of the MCT network, which consistently surpassed those of the SOTA methods. These results suggest that current single-level fusion strategies may have reached their theoretical limits, necessitating true multilevel multimodal information fusion in advancing LULC classification.

The proposed MCT framework not only addresses the performance bottlenecks of single-modal approaches, but also provides a generalizable and efficient architecture for future research in multimodal RS tasks. In the future, we will explore lightweight deployments for edge computing, semi-supervised adaptations for label-efficient scenarios, and applications beyond LULC classification, such as environmental monitoring and urban change detection.

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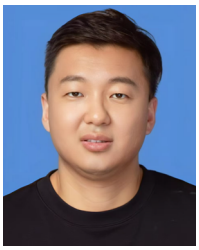
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